

DETAILED DESCRIPTION OF THE INVENTION

Regulatory Positioning and Intended Use Variants

In various embodiments, the penile anatomical geometry profiling system may be configured under differing intended use frameworks without altering the core sensing and reconstruction architecture described herein. The system may be positioned as a consumer wellness device, a research instrumentation platform, a clinical assessment tool, or a component within a regulated medical device ecosystem depending on deployment configuration and software activation parameters.

In a consumer wellness configuration, the system may generate structural geometry profiles for informational, personalization, calibration, or compatibility applications without asserting diagnostic or therapeutic claims. Output metrics may be presented as structural descriptors, proportionality indices, or generalized classification categories. Data processing and storage may occur locally on user-controlled devices with optional encrypted cloud synchronization.

In a research configuration, the system may be utilized to collect anonymized structural geometry datasets for population analysis, statistical modeling, or biomechanical study. Research mode may enable extended data export formats, higher-resolution sampling parameters, and compatibility with analytical software platforms used in academic or institutional environments.

In a clinical or medical configuration, the system may be integrated with professional oversight workflows and structured reporting mechanisms. Clinical activation may enable expanded reconstruction detail, longitudinal structural change tracking, and integration with secured institutional record systems. Regulatory classification, labeling, and documentation may vary depending on jurisdiction and intended application.

In certain embodiments, firmware or software controls may selectively enable or disable features associated with specific intended use categories. Device-level configuration profiles may govern available output formats, sampling density, comparative analytics depth, and data export permissions. Such configurability allows a single hardware platform to support multiple regulatory pathways without altering the underlying geometry acquisition methodology.

The regulatory positioning and intended use variations described herein do not limit the inventive concepts of acquiring synchronized time-of-flight radial distance data and longitudinal motion measurements to generate a user-specific penile anatomical geometry profile. The system may operate within diverse compliance frameworks while preserving the fundamental architecture disclosed.